Merit Education Group

Evaluating One-Variable Expressions

Algebra 1 · Pacing ID: EvalOneVar

Standard: 9.3.5.5

Standard: 9.3.5.7

Standard: 9.3.5.8

Lesson Overview

Objectives	• Evaluate one-variable expressions using substitution and GEMDAS. • Model precise substitution syntax with parentheses to preserve sign integrity.
Key Vocabulary	Evaluate, Substitute, Expression, Variable, Value
Skills Targeted	Substitution with parentheses; sign integrity; GEMDAS application post-substitution
Materials	Whiteboards & dry-erase markers (1 per student) · Tiered Worksheet Packets (T1/T2/T3) · Exit Tickets (pre-printed or half-sheet) · Prime Time Speed Lane number cards (laminated, 20–30 composites) · Timer/Projector · Teacher diagnostic circulation checklist

Phase 1: Activate & Explore 16 min

🔔 Bell Ringer

4 min

Display slide with header Whiteboards Ready! Students solve independently on whiteboards; hold up answers on signal. Purpose: reactivate order of operations and integer arithmetic without variable context to isolate the substitution step later.

Companion: See U1L05_EvalOneVar_BellRinger.md.

📖 Direct Instruction

6 min

1. Define Evaluate:

To determine the numerical value of an expression when specific values are assigned to the variables. Requires substitution — replacing each variable with its given value, enclosed in parentheses to preserve sign integrity.

2. Model (Board):

Evaluate $3x + 4$ when $x = -2$.
Step 1: Write expression: $3x + 4$
Step 2: Substitute with parentheses: $3(-2) + 4$
Step 3: Apply GEMDAS => $-6 + 4 = -2$

3. GEMDAS Reminder:

Substitution does not change operation priority. Always resolve Exponents before Multiplication/Division, then Addition/Subtraction. Parentheses from substitution are resolved via multiplication immediately after.

Precision Language: We say "substitute with parentheses" not "plug in." Every substituted value must be wrapped in parentheses — even positive numbers — to build the habit for when negatives appear. Reading $3(-2)$ as "three times negative two" reinforces that the parentheses indicate multiplication, not grouping.

🌍 Culture & Career Spotlight

6 min

Cultural Context

No culture slide displayed per pacing directive.

Career: Retail Ops Coordinator — Mall of America, Bloomington

Education: BS | Salary: $102,950/year
Retail Operations Coordinators use one-variable expressions to forecast inventory turnover, schedule staffing, and model promotional impact. Example: $R(d) = 14,500 - 250d$ where d = participating stores running a promotion. Precision in evaluation directly impacts operational efficiency and profit margins.

Phase 2: Practice & Apply 26 min

🧭 Guided Practice

8 min

Example 1 (Positive Substitution):
Evaluate $5m + 9$ when $m = 4$.

🧮 GEMDAS Checkpoint: Multiply first, then add.
Solution: $5(4) + 9 = 20 + 9 = 29$

Example 2 (Exponents & Coefficients):
Evaluate $2a^2 - 7$ when $a = 3$.

🧮 GEMDAS Checkpoint: Exponents before multiplication; substitute with parentheses.
Solution: $2(3)^2 - 7 = 2(9) - 7 = 18 - 7 = 11$

Example 3 (Negative Substitution):
Evaluate $10 - 3y$ when $y = -2$.

🧮 GEMDAS Checkpoint: Multiplication of negatives yields positive; subtraction becomes addition.
Solution: $10 - 3(-2) = 10 + 6 = 16$

GEMDAS Reminder: After substitution, the expression becomes a numerical computation. Apply GEMDAS strictly left to right for same-precedence operations. Exponents apply only to the base inside the parentheses, not the coefficient — $2a^2$ means $2(a^2)$, not $(2a)^2$.

📝 Independent Worksheet + Circulate & Support

18 min

Tiered Structure:

T1 Foundational	Linear expressions with positive integer substitutions (e.g., $4x - 3$ when $x = 5$).
T2 Application	Expressions with exponents and negative values requiring strict GEMDAS sequencing.
T3 Extension	Real-world scenarios requiring expression setup and evaluation (parking fee models, print shop costs).

Teacher Action: Circulate systematically with a diagnostic checklist. Target errors: missing parentheses, exponent scope confusion, sign mishandling during multiplication. First 9 minutes: silent independent work. Second 9 minutes: provide immediate corrective feedback using redirect questions.

Phase 3: Assess & Reflect 8 min

🎲 Game — Prime Time Speed Lane

5 min

Zero-Device Activity. Students form two parallel lines facing each other. Teacher calls a composite number (e.g., “12”, “15”, “20”). Pairs must rapidly state the prime factorization or product pair that clears the lane first. First team to clear 6 numbers wins. Reinforces multiplicative reasoning critical for efficient evaluation without calculators.

Zero-device activity — number cards pre-laminated.

🎫 Exit Ticket

2 min

Evaluate $4k - 5$ when $k = 3$. Show substitution step explicitly.
Success Criteria: Correct parentheses usage, accurate GEMDAS execution, final value of 7.

🔮 Today & Tomorrow

1 min

Today: We evaluated one-variable expressions with precision using substitution and GEMDAS.

Tomorrow: Lesson 6 — Evaluating Multi-Variable Expressions: We extend this skill to two-variable expressions and explore how changing multiple inputs affects output values.

SPED & ELL Cross-Reference

S

[**SPED**

Provide substitution scaffold mat with pre-printed parentheses slots. Allow calculator use for arithmetic only (not operation sequencing). Chunk worksheet into 3 sections with self-check answer keys.]

L

[**ELL**

Supply bilingual vocabulary anchor (evaluate = determine numerical value, substitute = replace). Use sentence frames: "When _____, I substitute _____ to get _____." Pair with strategic peer during Speed Lane game for language modeling.]

Misconceptions & Redirect Questions

	Misconception	**Redirect Question**
Corrective Strategy	Skipping parentheses during substitution (e.g., $3 - 2$ instead of $3(-2)$)	**"How does omitting parentheses change how we read the operation?"**
Require explicit bracket notation on all future work.	Applying exponent to coefficient ($(2x)^2$ vs $2x^2$)	**"Does the exponent touch the 2, or only what's directly attached?"**
Color-coding: base in blue, exponent applies only to blue region.	Subtracting a negative incorrectly ($10 - (-4) = 6$)	**"What operation is actually happening after you distribute the negative sign?"**
Model number line; emphasize $-(-)$ becomes $+$.		

Materials Checklist

- 🔲 Whiteboards & dry-erase markers (1 per student)
- 🔲 Exit Tickets (pre-printed or half-sheet)
- 🔲 Tiered Worksheet Packets (T1 / T2 / T3)
- 🔲 Prime Time Speed Lane number cards (laminated, 20–30 composites)

- 🔲 Timer/Projector for phase transitions
- 🔲 Teacher diagnostic circulation checklist
- 🔲 Substitution scaffold mat (SPED/ELL support)